

Domain-Specific RAG for Hugging Face Documentation

End-to-end knowledge indexing, grounded retrieval, answer generation, and quality evaluation

OFFLINE INDEXING PATH

One-time / incremental ingestion

HF Documentation

2,647 source docs
text + source path
validation & cleaning

Chunking

Sliding window
1,000 characters
200-character overlap

Lineage Metadata

27,434 chunks
chunk ID, source, offsets
document & chunk index

BGE Embeddings

Batched encoding
BAAI/bge-small-en-v1.5
384-dim, L2 normalized

Milvus Lite

Persistent vector store
FLOAT_VECTOR (384) • IP metric
batched insert + full ID read-back

27,434 verified stored records

ONLINE QUESTION-ANSWERING PATH

User Question

Natural language
e.g. fine-tune a model?

Query Embedding

BGE instruction prefix
same 384-dim space
L2 normalized

Bi-Encoder Retrieval

Milvus IP top-k search
ranked chunks + score
no cross-encoder reranking

Evidence Guardrail

Score threshold filtering
top 2 qualified chunks
reject insufficient evidence

Grounded Answer

Qwen2-1.5B-Instruct
context-only response
citations + diagnostics

Evaluation & Observability

Retrieval: Precision@k, Recall@k • Generation: Opik Answer Relevance & Hallucination • Per-case failure diagnosis